Part I — SIEM Fundamentals

1. What does SIEM stand for?
  Security Information and Event Management
2. What is the main role of a SIEM?
  To collect, analyze, and correlate security logs from different sources (firewall , IPS , IDS) in order to detect threats

3. What types of data does a SIEM collect?
 logs (networek , Application , host …) , network flow data(NetFlow and IPFIX) , threat inteligence (roge IPs, malicouse domines.., malware hash)
the methode of gathering data is :

1)push model ⇒ usually the agent is the responsible of sending data after install the agent at the endpoint he send the data in real time at the port 1514 to he manger
2) pull model (Agentless) ⇒ in somme case we conn't install Agent inside it like (switch , router ,firewall..) in te case we have to methodes :
    Agentless Monitoring ⇒ the manger connect to the device via ssh and execute somme commande to fetch data
    syslog ⇒ the external device send his data his data via syslog protcol at port 514

4. What is the difference between logs, events, and alerts?

Logs are the raw records generated by systems and devices that describe what happened.

An event is a meaningful occurrence derived from logs
An alert is a warning generated when an event or group of events indicates a potential threat.
5. Why is log correlation important?
because may log from a single system appear begin but when we correlate multiple source of logs we discover that is malicaouse

6. What is a false positive?
when the SIEM lunch something as suspicious but after the analysis we discover that is it begin

7. What is an incident in SIEM?

Incident is declared when the alert indicates real or likely malicious activity.

8 What is the difference between detection and prevention

Detection identifies and alerts on threats that occur, while prevention actively blocks or stops threats before they can cause harm

9. What are the main components of a SIEM architecture?

```
[Data Sources]
      ↓
[Collection]
      ↓
[Parsing & Normalization]
      ↓
[Storage]
      ↓
[Analysis Engine]
      ↓
[Alerting]
      ↓
[Dashboard + Response]
```

or in other architect somthing like that :

log source => colaction => aggregation , normlazation (parsing) => correlation => Alreting , analytics , ashbored

10. What is a use case in SIEM?

is detection logiuqe designed to convert raw data to security Alret
it consiste of (data source , detaction logique , Thredsholes , context , responses)
in other words a use cases is :

Use Case = Logs + Logic + Context

Wazuh provides three main ways to create use cases:

1. Rules: Detect suspicious activity.
2. Decoders: Parse specific log formats.
3. Alerts/Policies: Define what action to take when a rule triggers.

example of SSH BruteForce usecase

```
<!-- create a use case wich is a rule of detecting ssh brout force  -->
<group name="ssh_bruteForce">
        <rule id="100002" level="15" frequency="5" timeframe="600">
                <decoded_as>sshd</decoded_as>
                <if_matched_sid>5716</if_matched_sid>
                <same_source_ip />
                <description>SSH brute force Attack detected</description>
        </rule>
</group>
```

11.Give two examples of attacks detectable by a SIEM.

-Brute Force Attacks

-Malware infections

Part II — XDR vs SIEM

1. What does XDR stand for?

Extended Detection and Response

2. What is the main difference between SIEM and XDR? SIEM focuses only on collecting and analysing logs, whereas XDR focuses on detection and response across multiple sources in a coordinated and intelligent way

More precise wording:

> SIEM:*A system that collects and analyses data (logs) to generate alerts.*
>
> *XDR:A system that collects, correlates, analyses, and responds across multiple layers (Endpoint, Network, Cloud).*

3. What does a SIEM focus on?

SIEM focuses on collecting and analysing event logs from various systems to detect security threats.

4. What does an XDR focus on?

focus on Detection and Correlation and Response across multiple layers

5. Can SIEM and XDR work together? Explain briefly.

Yes, SIEM and XDR can work together.

Brief explanation:
SIEM collects and stores logs from many systems and provides centralized visibility, while XDR analyzes security data across endpoints, networks, and other layers to detect and respond to threats. When integrated, XDR feeds enriched, high-quality alerts into SIEM, and SIEM provides broader context and long-term log storage, improving overall detection and incident response.

6. What types of sources does XDR monitor?

XDR monitors multiple types of sources, including:

- *Endpoints (computers, servers, mobile devices)*
- *Network traffic (internal and external communications)*
- *Cloud environments (cloud services and workloads)*

- *Email systems (phishing and malicious attachments)*
- *Identity systems (user accounts, authentication, access logs)*

7. What is automated response in XDR? :

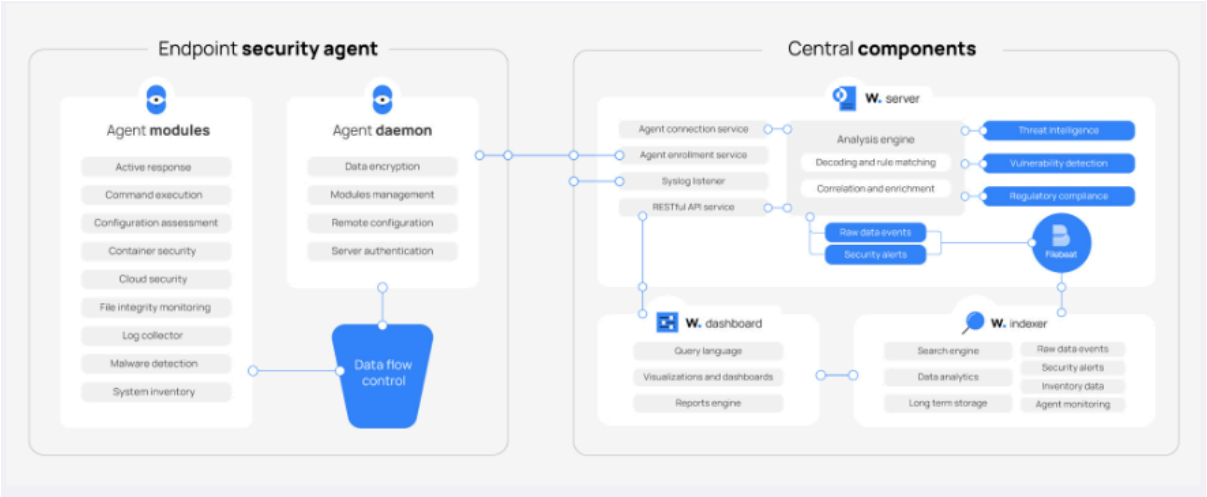
The system can automatically take actions to stop or contain a threat without human intervention.

examples (Isolating an infected endpoint ,Blocking a malicious IP address ,Disabling a compromised user account)

Part III — Wazuh Architecture

1. What is Wazuh and what are its main components?

Wazuh is an open-source security platform that provides host-based intrusion detection (HIDS), security monitoring, and compliance management. It helps organizations detect threats, monitor integrity, and respond to security incidents in real time. Wazuh is often used for log analysis, vulnerability detection, file integrity monitoring, and endpoint security.



2. Explain the role of each component.

1. Endpoint Security Agent

This is the software installed on the monitored devices (servers, workstations, or cloud instances). It is divided into:

- *Agent Modules: The active sensors that collect logs, detect malware, check configurations, and monitor file integrity.*
- *Agent Daemon: The background service that manages the modules, encrypts data, and handles communication with the central server.*
- *Data Flow Control: Regulates the flow of data being sent to prevent network overload.*

-. W. Server (Wazuh Server)

The central brain of the system. It receives data from the agents, decodes it, and runs it through an analysis engine to match it against security rules, detect vulnerabilities, and identify threats.

-. W. Indexer (Wazuh Indexer)

A highly scalable search and analytics engine. It securely stores the alerts generated by the Wazuh server and allows for fast querying, data analysis, and long-term storage. (Filebeat acts as the bridge that ships data from the Server to this Indexer).

-. W. Dashboard (Wazuh Dashboard)

The web user interface used to visualize and analyze the data. It provides dashboards, reporting engines, and search tools for security analysts to monitor the environment.

3. Describe the data flow from Agent to Dashboard.

Agent:

- *Installed on endpoints to collect logs, monitor file integrity, detect malware/rootkits, and track system events.*
- *Sends the data securely (TLS-encrypted) to the Manager.*

Manager:

- *Decodes raw logs into structured events.*
- *Applies rules to detect threats, policy violations, or anomalies.*
- *Correlates events from multiple agents to reduce false positives.*
- *Generates alerts with enriched context (like threat intelligence or endpoint info).*

Indexer (Wazuh Indexer):

- *Stores and indexes the alerts and event data.*

- *Enables fast searches, historical analysis, and reporting.*

API:

- *Exposes Manager's data so dashboards or automation tools can query alerts, agent status, and metrics.*

Dashboard (Wazuh App):

- *Visualizes alerts and events in real-time.*
- *Allows filtering, searching, and investigating incidents.*

- *Shows compliance and security metrics for the organization.*

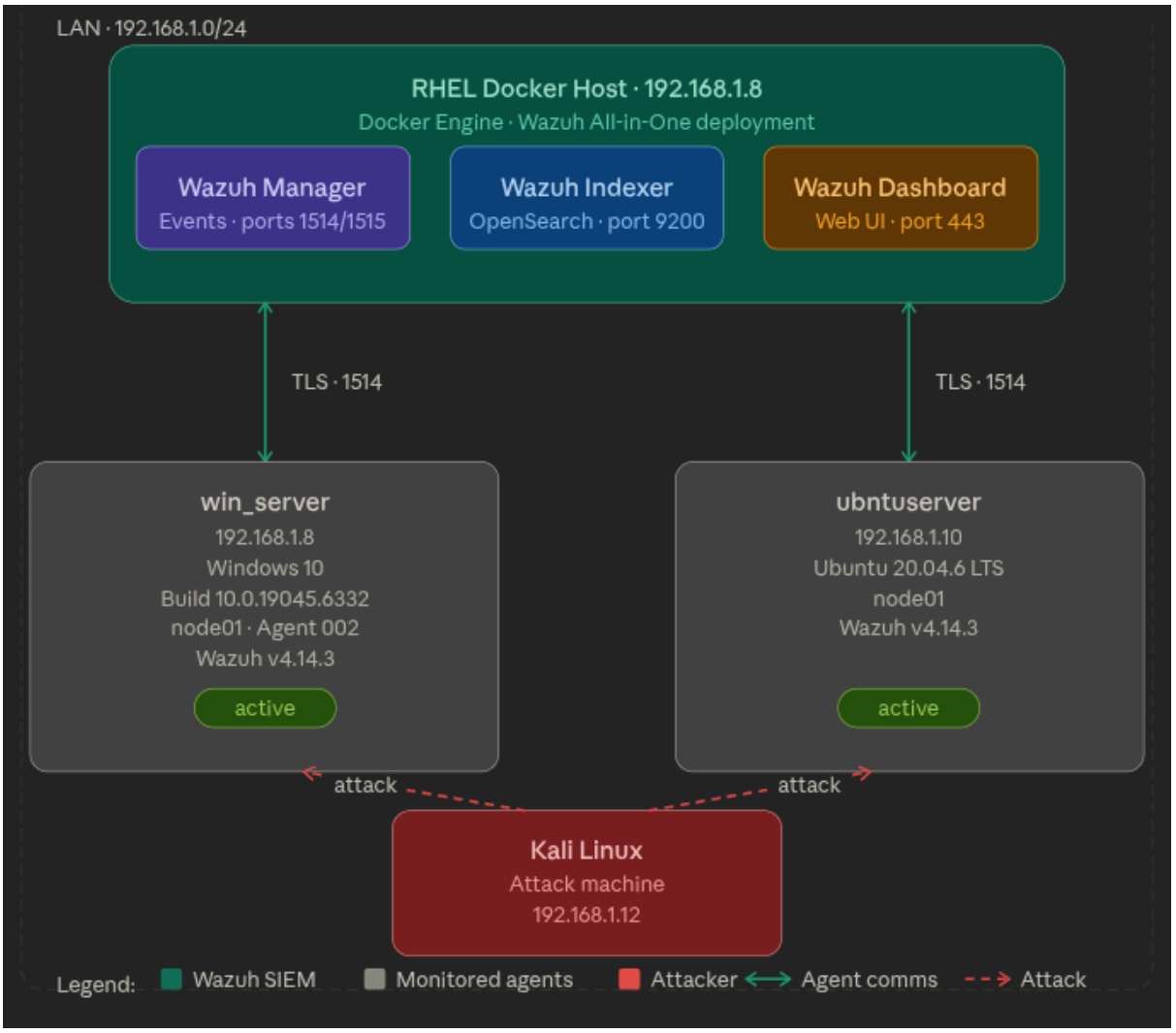
Part IV — Wazuh Installation

Wazuh Docker deployment

Single-node stack: This stack deploys one of each Wazuh central component as a separate container. It includes:

- Wazuh indexer container: Stores and indexes security data collected by the Wazuh manager.
- Wazuh manager container: Analyzes collected security events, applies detection rules, and manages Wazuh agents.
- Wazuh dashboard container: Centralized web interface for monitoring, searching, and
- managing Wazuh.

Lab 1 - SIEM/XDR- Wazuh Installation



Docker host requirements

1.set max_map_count

```
[ababil@aaliiwaamra ~]$ sysctl -w vm.max_map_count=262144
sysctl: permission denied on key "vm.max_map_count"
[ababil@aaliiwaamra ~]$ sudo !!
sudo sysctl -w vm.max_map_count=262144
[sudo] password for ababil:
vm.max_map_count = 262144
[ababil@aaliiwaamra ~]$
```

Lab 1 - SIEM/XDR- Wazuh Installation

Cloning the repository

1. Clone the Wazuh Docker repository to your system:

```
[ababil@aaliiwaamra ~]$ cd Downloads/
[ababil@aaliiwaamra Downloads]$ git clone https://github.com/wazuh/wazuh-docker.git -
b v4.14.3
Cloning into 'wazuh-docker'...
remote: Enumerating objects: 16481, done.
remote: Counting objects: 100% (519/519), done.
```

Wazuh self-signed certificates

```
Activities   Terminal                                   Mar 12 06:01
ababil@aaliiwaamra:~/Downloads/wazuh-docker/single-node
[ababil@aaliiwaamra Downloads]$ ls
config.yml  wazuh-certificates-tool.log  wazuh-certs-tool.sh  wazuh-docker
[ababil@aaliiwaamra Downloads]$ cd wazuh-docker/
[ababil@aaliiwaamra wazuh-docker]$ cd single-node/
[ababil@aaliiwaamra single-node]$ docker compose -f generate-indexer-certs.yml run --rm generator
failed to connect to the docker API at unix:///var/run/docker.sock; check if the path is correct and if the daemon is running: dial unix /var/run/docker.sock: connect: no such file or directory
[ababil@aaliiwaamra single-node]$ ls
config  docker-compose.yml  generate-indexer-certs.yml  README.md
[ababil@aaliiwaamra single-node]$ sudo systemctl status docker
[sudo] password for ababil:
○ docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: disabl
     Active: inactive (dead)
TriggeredBy: ○ docker.socket
       Docs: https://docs.docker.com
[ababil@aaliiwaamra single-node]$ sudo systemctl start docker
[ababil@aaliiwaamra single-node]$ docker compose -f generate-indexer-certs.yml run --rm generator
permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
[ababil@aaliiwaamra single-node]$ sudo !!
sudo docker compose -f generate-indexer-certs.yml run --rm generator
[+] 1/1te 1/1
 ✔ Network single-node_default Created                                    0.4s
Image wazuh/wazuh-certs-generator:0.0.4 Pulling
cf48939c4ae7 Pulling fs layer 0B
f0d8a57b0a96 Pulling fs layer 0B
035f2e5bf75c Pulling fs layer 0B
df36708a1273 Pulling fs layer 0B
bc8de1f101c4 Download complete 0B
df36708a1273 Download complete 0B
cf48939c4ae7 Download complete 0B
f0d8a57b0a96 Downloading 1.049MB
f0d8a57b0a96 Downloading 1.049MB
f0d8a57b0a96 Downloading 1.049MB
035f2e5bf75c Downloading 1.049MB
035f2e5bf75c Downloading 1.049MB
f0d8a57b0a96 Downloading 1.049MB
f0d8a57b0a96 Downloading 2.097MB
035f2e5bf75c Downloading 1.049MB
f0d8a57b0a96 Downloading 2.097MB
035f2e5bf75c Downloading 1.049MB
035f2e5bf75c Downloading 1.049MB
f0d8a57b0a96 Downloading 2.097MB
f0d8a57b0a96 Downloading 3.146MB
035f2e5bf75c Downloading 2.097MB
f0d8a57b0a96 Downloading 3.146MB
035f2e5bf75c Downloading 2.097MB
f0d8a57b0a96 Downloading 3.146MB
035f2e5bf75c Downloading 2.097MB
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
```

```
Image wazuh/wazuh-certs-generator:0.0.4 Pulled
Container single-node-generator-run-db50f04f5cb3 Creating
Container single-node-generator-run-db50f04f5cb3 Created
Checking https://packages.wazuh.com/4.14/wazuh-certs-tool.sh ...
Downloaded wazuh-certs-tool.sh from https://packages.wazuh.com/4.14/
12/03/2026 05:00:56 INFO: Verbose logging redirected to //wazuh-certificates-tool.log
12/03/2026 05:00:56 INFO: Generating the root certificate.
12/03/2026 05:00:57 INFO: Generating Admin certificates.
12/03/2026 05:00:57 INFO: Admin certificates created.
12/03/2026 05:00:57 INFO: Generating Wazuh indexer certificates.
12/03/2026 05:00:58 INFO: Wazuh indexer certificates created.
12/03/2026 05:00:58 INFO: Generating Filebeat certificates.
12/03/2026 05:00:59 INFO: Wazuh Filebeat certificates created.
12/03/2026 05:00:59 INFO: Generating Wazuh dashboard certificates.
12/03/2026 05:01:00 INFO: Wazuh dashboard certificates created.
Moving created certificates to the destination directory
Changing certificate permissions
Setting UID indexer and dashboard
Setting UID for wazuh manager and worker
[ababil@aaliiwaamra single-node]$
```

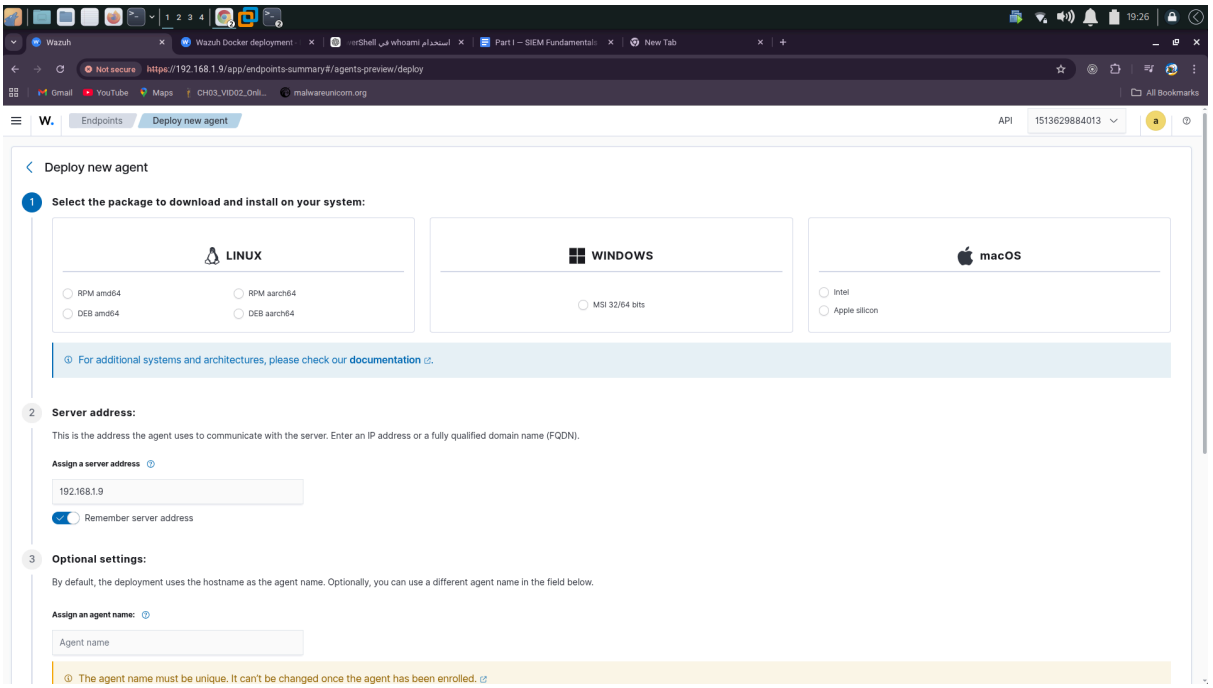
Lab 1 - SIEM/XDR- Wazuh Installation

wazuh.

Start the Wazuh Docker deployment using the docker compose command:

```
12/03/2026 05:00:58 INFO: Generating Filebeat certificates.
12/03/2026 05:00:59 INFO: Wazuh Filebeat certificates created.
12/03/2026 05:00:59 INFO: Generating Wazuh dashboard certificates.
12/03/2026 05:01:00 INFO: Wazuh dashboard certificates created.
Moving created certificates to the destination directory
Changing certificate permissions
Setting UID indexer and dashboard
Setting UID for wazuh manager and worker
[ababil@aaliiwaamra single-node]$ docker compose up -d
unable to get image 'wazuh/wazuh-indexer:4.14.3': permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
[ababil@aaliiwaamra single-node]$ sudo !!
sudo docker compose up -d
[+] up 42/45
[+] up 42/45uh/wazuh-indexer:4.14.3                                    Pulled                441.1s
[+] up 61/62uh/wazuh-indexer:4.14.3                                    Pulled                441.1s
 ✔ Image wazuh/wazuh-indexer:4.14.3          Pulled                                        441.1s
 ✔ Image wazuh/wazuh-dashboard:4.14.3        Pulled                                        440.9s
 ✔ Image wazuh/wazuh-manager:4.14.3          Pulled                                        495.0s
 ✔ Volume single-node_wazuh_active_response  Created                                       0.0s
 ✔ Volume single-node_wazuh-indexer-data     Created                                       0.0s
 ✔ Volume single-node_wazuh-dashboard-config Created                                       0.0s
 ✔ Volume single-node_wazuh_api_configuration Created                                      0.0s
 ✔ Volume single-node_wazuh_logs             Created                                       0.0s
 ✔ Volume single-node_wazuh_queue            Created                                       0.0s
 ✔ Volume single-node_wazuh_agentless        Created                                       0.0s
 ✔ Volume single-node_wazuh_wodles           Created                                       0.0s
 ✔ Volume single-node_wazuh_var_multigroups  Created                                       0.0s
 ✔ Volume single-node_filebeat_etc           Created                                       0.0s
 ✔ Volume single-node_filebeat_var           Created                                       0.0s
 ✔ Volume single-node_wazuh-dashboard-custom Created                                       0.0s
 ✔ Volume single-node_wazuh_etc              Created                                       0.0s
 ✔ Volume single-node_wazuh_integrations     Created                                       0.0s
 ✔ Container single-node-wazuh.manager-1     Started                                       5.5s
```

Deploy new agent

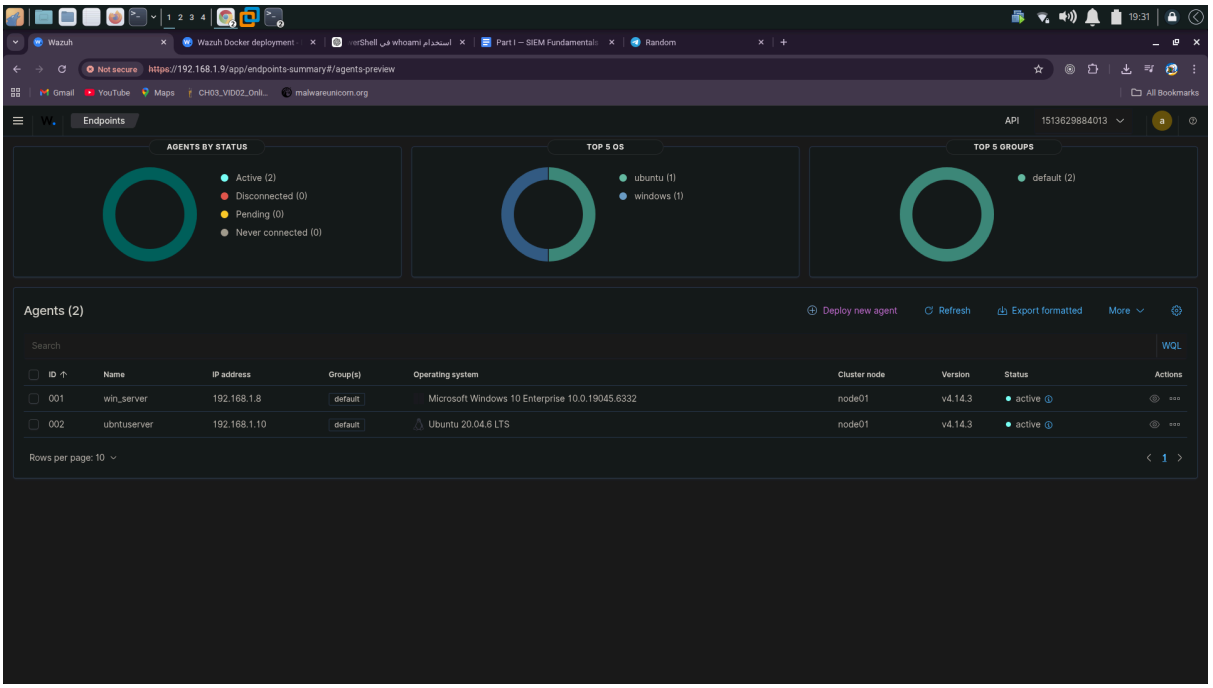
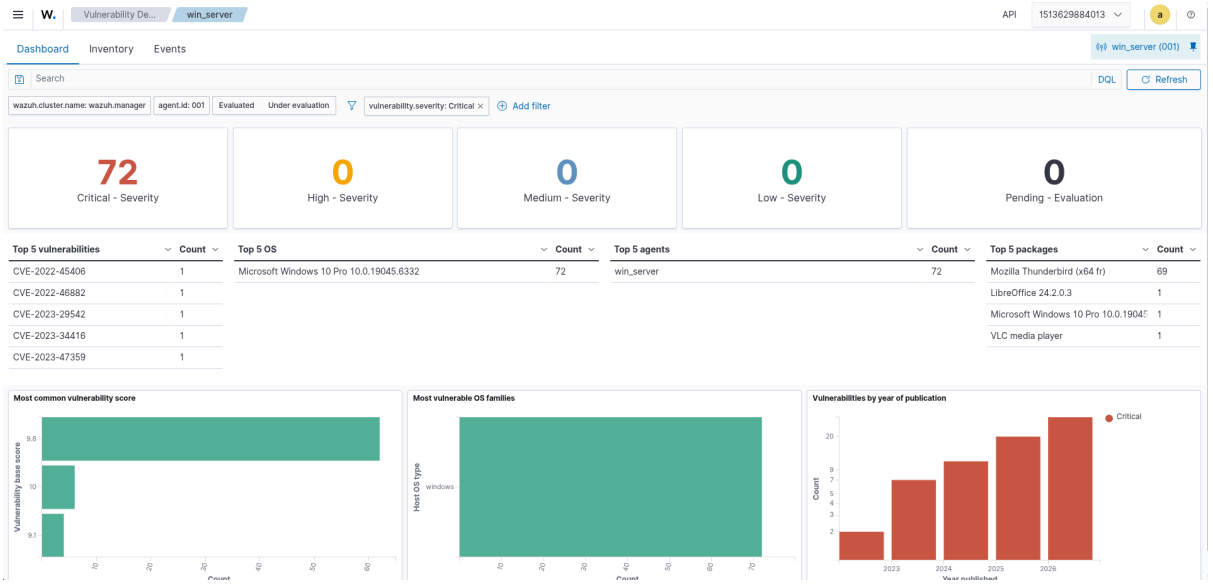


1st ubuntu Endpoint agent enrollment

Lab 1 - SIEM/XDR- Wazuh Installation

```
Last login: Thu Mar 12 05:28:40 2026
ababil@ubntumacine:~$ wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.3-1_amd64.deb
&& sudo WAZUH_MANAGER='192.169.1.10' WAZUH_AGENT_NAME='ubntumacine' dpkg -i ./wazuh-agent_4.14.3-1_amd64.deb
--2026-03-12 05:37:07--  https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.3-1_amd64.deb
Resolving packages.wazuh.com (packages.wazuh.com)... 3.160.188.113, 3.160.188.9, 3.160.188.85, ...
Connecting to packages.wazuh.com (packages.wazuh.com)|3.160.188.113|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13213254 (13M) [application/vnd.debian.binary-package]
Saving to: 'wazuh-agent_4.14.3-1_amd64.deb'

wazuh-agent_4.14.3-1_amd64.de 100%[===============================================>]  12.60M  6.73MB/s    in 1.9s

2026-03-12 05:37:09 (6.73 MB/s) - 'wazuh-agent_4.14.3-1_amd64.deb' saved [13213254/13213254]

Selecting previously unselected package wazuh-agent.
(Reading database ... 93992 files and directories currently installed.)
Preparing to unpack .../wazuh-agent_4.14.3-1_amd64.deb ...
Unpacking wazuh-agent (4.14.3-1) ...
Setting up wazuh-agent (4.14.3-1) ...
Processing triggers for systemd (245.4-4ubuntu3.24) ...
ababil@ubntumacine:~$ sudo systemctl daemon-reload
ababil@ubntumacine:~$ sudo systemctl enable wazuh-agent
Created symlink /etc/systemd/system/multi-user.target.wants/wazuh-agent.service → /lib/systemd/system/wazuh-agent.ser
vice.
ababil@ubntumacine:~$ sudo systemctl start wazuh-agent
ababil@ubntumacine:~$
```

2nd windows Endpoint agent enrollment

```
Administrateur : Windows PowerShell
 $env:tmp\wazuh-agent; msiexec.exe /i $env:tmp\wazuh-agent /q WAZUH_MANAGER='192.168.1.9' WAZUH_AGENT_GROUP='default' WA
ZUH_AGENT_NAME='win_server'
PS C:\Users\utilisateur>
PS C:\Users\utilisateur>
PS C:\Users\utilisateur> NET START Wazuh
Le service Wazuh démarre.
Le service Wazuh a démarré.

PS C:\Users\utilisateur>
```

Lab 1 - SIEM/XDR- Wazuh Installation